ModelIt!

Teacher's Guide

How Does the Brain Create Consciousness?

Modeling Neural Correlates, Emergence, and the Architecture of Subjective Experience

Quick Reference

Grade Level	Grade 12L3
NGSS Standard	HS-LS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-2

Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a multi-component model of neural correlates of consciousness, testing predictions from competing neuroscientific theories about which brain processes are necessary and sufficient for conscious experience.
Disciplinary Core Idea	**LS1.A Structure and Function** — Students analyze how the hierarchical organization of neural systems — from brainstem arousal through thalamic gating to cortical integration — produces the specific functions associated with different aspects of conscious experience.
Crosscutting Concept	**Structure and Function** — Students investigate how the structural connectivity of neural circuits (thalamo-cortical loops, prefrontal integration hubs, gamma synchrony networks) relates to the functional emergence of consciousness.

CAST Testing Alignment

MULTIPLE CHOICE:

A patient in a vegetative state shows preserved brainstem reflexes and sleep-wake cycles (Arousal State: present) but no purposeful behavior or communication. Based on the consciousness model, which disrupted component most likely explains their condition?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain how dreaming demonstrates that consciousness does not require external sensory input and identify which neural components sustain conscious experience during REM sleep. Reference at least three model components and their specific roles in generating dream consciousness.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Consciousness is the most familiar phenomenon in the universe — you experience it every waking moment — and simultaneously the most mysterious. Despite extraordinary advances in neuroscience, we still cannot explain how 86 billion neurons firing electrochemical signals produce the subjective experience of seeing color, feeling pain, or thinking a thought. The neural correlates of consciousness are increasingly well-mapped, but the explanatory gap between brain activity and subjective experience remains the deepest unsolved problem in science.

The Two Dimensions of Consciousness

Consciousness has two independent dimensions: arousal (wakefulness level) and content (what you are conscious of). The brainstem reticular activating system controls arousal — damage here causes coma. The cortex generates content — damage to visual cortex eliminates visual consciousness while leaving other modalities intact. These dimensions dissociate in interesting ways: coma is low arousal with no content, vegetative state is arousal without content, dreaming is moderate arousal with internally generated content, and normal waking is high arousal with external and internal content. This two-dimensional framework organizes the clinical disorders of consciousness and identifies the distinct neural systems that must both function for full conscious experience.

Global Neuronal Workspace Theory

Stanislas Dehaene and Jean-Pierre Changeux proposed that consciousness arises when sensory information, initially processed in local cortical circuits below awareness, ignites a network of long-range connections involving prefrontal and parietal cortices. This ignition broadcasts the information to all brain systems simultaneously — making it available for verbal report, decision-making, memory encoding, and action planning. The theory makes testable predictions: conscious perception should be all-or-none (information either ignites the workspace or does not), should involve a specific temporal signature (a late widespread cortical activation called the P3b wave), and should require intact prefrontal connectivity. These predictions have been largely confirmed experimentally.

Integrated Information Theory from Neuroscience

While IIT is a general theory of consciousness applicable to any system, its neuroscience predictions are specific: consciousness should correlate with the degree of information integration in cortical networks. The posterior cortex (including visual, auditory, and somatosensory areas) has the highest intrinsic connectivity and information integration in the brain, and IIT predicts this is the hot zone of consciousness — the minimal neural substrate for conscious experience. This prediction contrasts with Global Workspace Theory, which emphasizes prefrontal involvement. Distinguishing between these theories experimentally is an active area of research, with adversarial collaborations between theory proponents conducting pre-registered studies.

Clinical Disorders of Consciousness

The neuroscience of consciousness has direct clinical impact. Brain-injured patients may exist in coma (no arousal or awareness), vegetative state (arousal without awareness), minimally conscious state (intermittent awareness), or locked-in syndrome (full consciousness with no motor output except eye movements). Critically, research using fMRI has shown that 15-20% of apparently vegetative patients show brain responses to commands (imagine playing tennis) identical to healthy conscious volunteers. These covert consciousness patients are aware but trapped — unable to communicate through any behavioral channel. Detecting their consciousness requires the neuroscience principles modeled in this lesson and has profound ethical implications for their care.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten key components of the consciousness system: sensory cortex activation, arousal state, external stimulus complexity, thalamo-cortical loops, prefrontal integration, gamma synchrony, default mode network, information integration, recurrent processing, and conscious experience index.

E **Establish Relationships**

Students map relationships showing how arousal gates all cortical processing, how thalamo-cortical loops and recurrent processing are necessary for conscious access, and how prefrontal integration creates the global workspace.

V **Visualize & Model**

Students build a ten-component computational model of the neural correlates of consciousness, with distinct pathways for arousal, content, integration, and self-referential processing.

E **Evaluate Outcomes**

Students run full awareness, anesthesia, and dreaming scenarios to test which neural components are necessary for consciousness and what happens when specific components are disabled.

R **Revise & Extend**

Students add interoceptive processing, attention network control, or neurochemical modulation to explore how embodiment, selective attention, and brain chemistry shape the character of conscious experience.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Right now, you are conscious. You know you exist. You see colors, hear sounds, feel emotions. But WHY? 86 billion neurons are just cells firing chemicals. How do electrochemical signals become the experience of being you? This is the hardest question in all of science."*

Watch For

Show cover image. Moment of silence: ask students to spend 10 seconds simply noticing their own consciousness — the experience of experience itself. Then ask: what is it that you just noticed? Can you describe it?

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling the most mysterious phenomenon in the known universe: your own consciousness. You will map the neural systems that correlate with awareness and confront the hard problem — why physical brain activity produces subjective experience at all."*

Watch For

Have students read objectives. Pre-teach 'neural correlates' vs. 'neural causes' — the difference between brain patterns that accompany consciousness and brain processes that produce it. This distinction is the key conceptual foundation.

Slide 3: Big Question

3 min

Say: *"How does the physical brain create the non-physical experience of consciousness? Notice the word 'how' — we are not asking IF it does (it clearly does) but HOW the transition from neurons to experience occurs. This question has defeated every philosopher and neuroscientist for 2,500 years."*

Watch For

Quick-write: Students attempt to define consciousness without using the word 'conscious.' Share out and discuss: why is this so difficult? The difficulty of defining consciousness reflects the difficulty of explaining it.

Slide 4: LEVER Framework

2 min

Say: *"We will Label the neural components associated with consciousness, Establish their relationships based on neuroscience evidence, Visualize a model that tests consciousness theories, Evaluate three states of consciousness, and Refine by adding embodiment and neurochemistry."*

Watch For

Preview each LEVER step. Set expectations: this model maps correlates of consciousness, not consciousness itself. The gap between our model and actual experience is scientifically important — it IS the hard problem.

Slide 5: Activity 1: Components

8 min

Say: *"Three inputs set the stage for consciousness: sensory activation provides content, arousal provides the baseline brain state, and stimulus complexity determines what reaches awareness. Seven internal components represent the neural processes that may generate or at least accompany conscious experience."*

Watch For

Students sort external versus internal components. Discuss: why is Arousal State external? Because it is set by the brainstem independently of cortical content — you can be aroused (awake) without being conscious of anything specific.

Slide 6: Activity 2: Connections

8 min

Say: *"Arousal gates everything — without it, no other component can produce consciousness. Thalamo-cortical loops relay and filter sensory information. Prefrontal integration creates the global workspace. Gamma synchrony binds features together. But which connections are necessary for consciousness and which are merely typical?"*

Watch For

Students draw arrows showing relationships. Key discussion: identify the pathway from sensory input through thalamic relay through cortical processing to prefrontal integration. This is the proposed pathway from unconscious processing to conscious awareness.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run three states of consciousness: full waking awareness, anesthesia-induced unconsciousness, and dreaming. Watch which components change between states. The differences will reveal what consciousness requires."*

Watch For

Students run all three scenarios and create a comparison table. Key question: dreaming has high conscious experience with no external input — what does this prove about consciousness and sensory experience? It proves the brain can generate consciousness entirely internally.

Slide 8: Discoveries

5 min

Say: *"Your model reveals that consciousness requires both arousal AND cortical integration — it is not enough to have one without the other. It also reveals that consciousness does not require external input, that thalamo-cortical loops are a necessary gateway, and that the hard problem remains: we can map every correlate without explaining the experience."*

Watch For

Class discussion of discoveries. Ask the deep question: does our model explain consciousness, or does it describe the conditions under which consciousness occurs? Is there a difference? This distinction between explanation and description is the heart of the hard problem.

Slide 9: STEM Challenge

12 min

Say: *"A brain-injured patient appears to be in a vegetative state. But research shows 15-20% of such patients have hidden awareness. Design a protocol to detect covert consciousness using the neuroscience principles from your model. And then face the ethical question: what changes if you find consciousness?"*

Watch For

Groups design assessment protocols specifying neuroimaging tests, expected results for conscious versus unconscious patients, and ethical guidelines. Present and evaluate: which protocol is most sensitive? Most ethical? What happens when the result is uncertain?

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Sensory Cortex Activation, Arousal State, External Stimulus Complexity
- **INTERNAL:** Thalamo-Cortical Loop Activity, Prefrontal Integration Level, Gamma Synchrony Index, Default Mode Network Activity, Information Integration (Phi), Recurrent Processing Depth, Conscious Experience Index

Relationships

- **Arousal State to Thalamo-Cortical Loop Activity:** POSITIVE (+) — Arousal systems in the brainstem directly modulate thalamic relay neurons — higher arousal opens thalamic gates, enabling sensory information to reach the cortex. Under anesthesia, reduced arousal disrupts thalamo-cortical loops, which is a primary mechanism of consciousness loss.
- **Thalamo-Cortical Loop Activity to Prefrontal Integration Level:** POSITIVE (+) — Active thalamo-cortical loops relay processed sensory information to prefrontal cortex for integration into the global workspace. When loops are disrupted, information remains in local circuits without reaching prefrontal integration or conscious access.
- **Prefrontal Integration Level to Gamma Synchrony Index:** POSITIVE (+) — Prefrontal activation orchestrates long-range gamma synchrony across cortical areas, binding separately processed features into unified conscious percepts. Higher prefrontal engagement produces more widespread gamma coordination.
- **Sensory Cortex Activation to Thalamo-Cortical Loop Activity:** POSITIVE (+) — Stronger sensory cortex activation drives more robust feedback to the thalamus, strengthening the recurrent loop. However, sensory activation alone without thalamic relay does not produce conscious experience, as demonstrated in blindsight.
- **Default Mode Network Activity to Conscious Experience Index:** POSITIVE (+) — Default mode network activation contributes the self-referential dimension of consciousness — the sense of being a continuous self with a narrative history. During dreaming, default mode activity drives vivid internal experience even without external input.
- **Recurrent Processing Depth to Information Integration (Phi):** POSITIVE (+) — Deeper recurrent processing — where feedback signals from higher cortical areas modulate lower areas — increases the integration of information across the network, raising phi. Purely feedforward processing generates low phi and no conscious experience.
- **Information Integration (Phi) to Conscious Experience Index:** POSITIVE (+) — According to IIT, phi directly corresponds to the richness of conscious experience. Higher phi means the system generates more information as an integrated whole, which the theory identifies with more complex and vivid experience.
- **Gamma Synchrony Index to Conscious Experience Index:** POSITIVE (+) — Gamma synchrony binds distributed neural representations into coherent percepts. Higher synchrony correlates with clearer, more vivid conscious experiences, while disrupted synchrony produces fragmented or absent awareness.

Key Discoveries

- Consciousness requires both arousal (the brainstem setting cortical excitability) and content (cortical processing of information) — the model reveals these as independent dimensions, explaining why coma (no arousal) and vegetative states (arousal without content) are both states of impaired consciousness but through different mechanisms
- Thalamo-cortical loops and recurrent processing appear to be necessary conditions for consciousness — purely feedforward processing (sensory cortex activation without recurrence) produces neural responses without conscious experience, as demonstrated in blindsight and subliminal perception
- The default mode network enables self-referential consciousness even without external stimulation — dreaming and mind-wandering are explained by default mode activation generating internal content that enters the global workspace through the same prefrontal integration pathway used for external stimuli

- Gamma synchrony may serve as the binding mechanism that unifies separately processed features into coherent conscious percepts — when synchrony is disrupted by anesthesia or brain damage, patients report fragmented or absent experience even when individual sensory systems function normally

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Consciousness is located in a specific brain region"	No single brain region is the seat of consciousness. Consciousness emerges from the dynamic interaction of multiple systems — brainstem arousal, thalamic relay, cortical processing, prefrontal integration, and gamma synchrony. Damage to any of these systems impairs consciousness in different ways. The posterior cortex may be the hot zone for conscious content, but it requires arousal from the brainstem and integration from the prefrontal cortex to function. Consciousness is a network property, not a regional property.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If you are not responding, you are not conscious"	Up to 15-20% of patients diagnosed as vegetative based on behavioral assessment show brain evidence of conscious awareness when tested with neuroimaging. These covert consciousness patients can follow commands mentally (imagine playing tennis, imagine walking through your house) even though they cannot produce any motor output. Behavioral unresponsiveness tells you about the motor system, not necessarily about consciousness.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We understand how the brain creates consciousness"	We have made extraordinary progress mapping the neural correlates of consciousness — the brain patterns that consistently accompany conscious experience. But correlation is not causation, and even complete correlation does not explain the mechanism. We can predict WHEN someone will be conscious based on brain activity patterns, but we cannot explain WHY certain patterns of neural firing produce subjective experience while others do not. The hard problem of consciousness — the explanatory gap between physical processes and felt experience — remains unsolved. Intellectual honesty requires acknowledging this gap while continuing to make progress on what we can study.